DUBAI HOUSE PRICE

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Executive Summary

- This project delivers a comprehensive real estate analytics dashboard designed to support strategic decision-making in Dubai’s property market.
- The dashboard provides:
- Market pricing trends
- Location-based performance
- Property demand analysis
- Developer performance insights
- It enables stakeholders to make data-backed investment decisions

Objective

- The main objectives of this project are:
- To analyze property price distribution across different locations.
- To identify high-demand property types and bedroom configurations.
- To evaluate price per square foot for better investment decisions.
- To study developer performance and market share.
- To build an interactive dashboard for data-driven decision-making.

Problem Statement

- The Dubai real estate market is vast and dynamic, with thousands of property listings across different locations, property types, and price ranges.
- However:
- Investors struggle to identify profitable locations.
- Buyers find it difficult to compare property values.
- Agents lack a clear view of demand trends.
- Raw data alone does not provide meaningful insights.
- There is a need for an interactive and analytical dashboard that transforms raw housing data into clear, actionable insights for better decision-making

Dataset Overview

The dataset consists of Dubai housing property listings.

Key Attributes:

- Price – Selling price (AED)
- Bedrooms – Number of bedrooms
- Bathrooms – Number of bathrooms
- Size (Sqft) – Total property area
- Property Type – Apartment, Villa, Townhouse
- Year Built – Construction year

The dataset contains thousands of property records used for analysis.

Data Preparation & Cleaning

Steps Performed:

- Removed duplicate records
- Handled missing values
- Corrected inconsistent text formats
- Converted price and size columns to numeric format
- Verified data accuracy

Calculated Fields Created:

- Price per Sqft = Price ÷ Size
- Property Age = 2025 – Year Built
- Listing Category = Budget / Mid-Range / High-End.

Key Performance Indicators (KPIs)

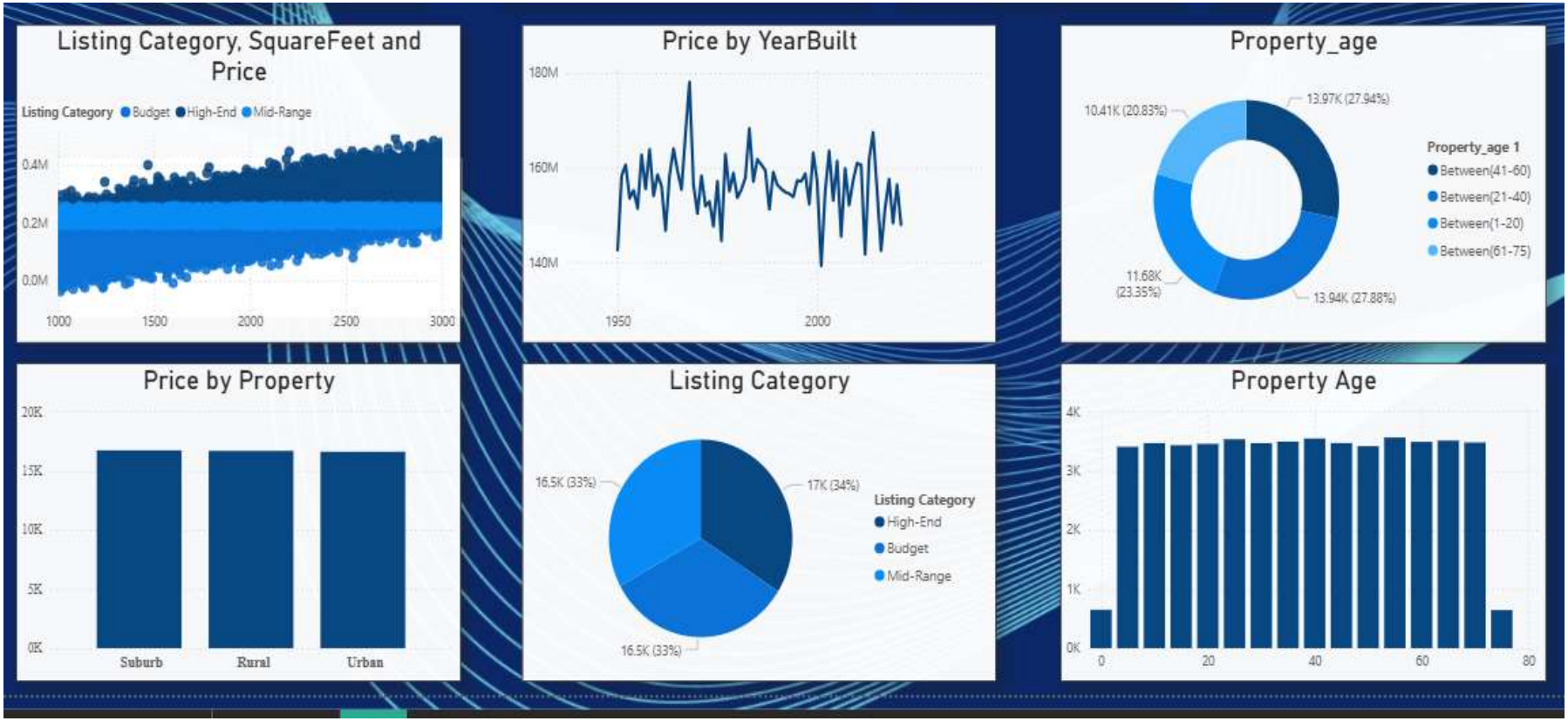
- Total Listings
- Average Price
- Average Size (Sqft)
- Highest Priced Property
- Average Price per Sqft
- These KPIs provide a quick summary of the real estate market.

Dashboard Visualizations

- The dashboard includes:
- Bar Chart – Price by property type
- Stacked Column Chart – Listings by bedrooms
- Line Chart – Price trend over time
- Pie Chart-Listing Category
- Donet Chart- Property Age

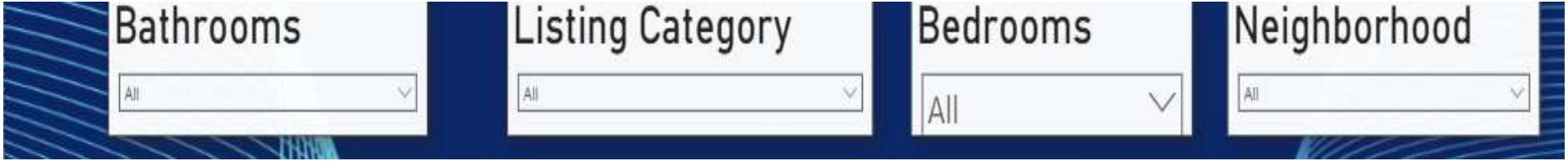
Each visualization helps in understanding market trends clearly.

output


Listing Category, SquareFeet and Price
Listing Category Budget High-End Mid-Range
0.4M
0.2M
0.0M
1000
1500
2000
2500
3000
Price by YearBuilt
180M
160M
140M
1950
2000
Property_age
10.41K (20.83%)
13.97K (27.94%)
11.68K (23.35%)
13.94K (27.88%)
Property_age 1
Between(41-60)
Between(21-40)
Between(1-20)
Between(61-75)
Price by Property
20K
15K
10K
5K
0K
Suburb
Rural
Urban
Listing Category
16.5K (33%)
17K (34%)
16.5K (33%)
Listing Category
High-End
Budget
Mid-Range
Property Age
4K
3K
2K
1K
0K
0
20
40
60
80

Interactivity Features

- To improve user experience, interactive elements were added:
- Slicers (Bedrooms, Property Type, Location, Price Range)
- Dynamic filtering
- Drill-through for detailed property view
- Tooltips showing price per sqft
- These features allow users to explore the data easily.



Insights & Findings

- Key findings from the analysis:
- Premium locations have higher average prices.
- Apartments dominate the market in total listings.
- Mid-range properties show stable demand.
- Price per sqft varies significantly by location.
- Certain developers dominate the high-end segment.

Business Recommendations

- Based on the analysis:
- Invest in mid-range properties for steady returns.
- Focus on high-growth areas for long-term gains.
- Compare price per sqft before purchasing.
- Partner with top-performing developers.
- Monitor location-based trends before investing.

Conclusion

- The Dubai Real Estate Dashboard successfully transforms raw data into actionable insights.
- It helps investors and agents understand pricing patterns, demand trends, and developer performance.
- This project demonstrates how data analytics and Power BI can support smart real estate decisions.